

Software Requirements Specification (SRS)

Inventory, Production, Sales & Accounting Management System

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the Inventory, Production, Sales, Repairs, Accounts, and Financial Management System.

The purpose of this system is to manage:

- Inventory and stock operations
- Production and recipe management
- Supplier and customer records
- Purchase, sales, and repair invoices
- Accounts and payments
- Expenses and stock adjustments
- Reporting and statements
- User management

The system will help businesses manage inverter manufacturing, inventory tracking, supplier/customer balances, production costs, repairs, and overall financial operations from a centralized platform.

1.2 Scope

The application will provide complete operational management for inverter manufacturing and sales businesses.

The system includes:

- Dashboard analytics
- Accounts management
- Supplier and customer management
- Item and inventory tracking
- Recipe and production management
- Purchase, sales, and repair invoice management
- Supplier/customer payments
- Expense management
- Stock adjustment operations
- User management
- CSV/PDF export features
- Search, filters, and pagination

The system must support accurate stock valuation, account balance tracking, invoice generation, production costing, and detailed financial reporting.

1.3 Intended Users

The system will be used by:

- Business owners
 - Account managers
 - Inventory managers
 - Production managers
 - Sales staff
 - Repair department staff
 - Administrative users
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1.4 Definitions

Term	Description
Current Balance	Real-time balance after all transactions
Opening Balance	Initial balance entered manually
Recipe	Formula of items required to manufacture an inverter
Production Batch	Manufacturing record of produced inverters
FOC	Free of Cost repair
Outstanding Balance	Remaining payable or receivable amount
CSV Export	Download data in CSV format
PDF Export	Download reports or statements in PDF format

2. Overall System Description

2.1 System Features

The system shall include: 1. Dashboard 2. Accounts Management 3. Supplier Management 4. Customer Management 5. Item & Inventory Management 6. Category Management 7. Recipe Management 8. Production Management 9. Purchase Invoice Management 10. Sales Invoice Management 11. Repair Invoice Management 12. Sold Inverters Tracking 13. Supplier Payments 14. Customer Payments 15. Stock Adjustments 16. Expense Categories 17. Expense Management 18. User Management

2.2 General Functional Requirements

2.2.1 Search Functionality

- Every search input must search data across all columns.
- Search results must update dynamically.

2.2.2 Sorting

- Alphabetical lists must sort from A-Z.
- Transaction-based tables must sort in descending date order.
- Latest records must appear first.

2.2.3 Export Features

The system shall support:

- CSV export
- PDF download

2.2.4 Pagination

All major tables must support pagination.

2.2.5 CRUD Operations

The system shall support:

- Create
 - Read/View
 - Update/Edit
 - Delete
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3. Functional Requirements

3.1 Dashboard Module

Description

The dashboard shall display summary statistics and financial totals.

Functional Requirements

Dashboard Cards

The dashboard shall display: 1. Total Purchase Cost 2. Total Expenses Cost 3. Total In-Stock Amount 4. Total Sale Price 5. Total Repair Cost 6. Total Sold Inverters Profit 7. Total Amount To Paid 8. Total Amount To Received 9. Total Current Balance 10. Total Production Cost 11. Overall Profit

Redirection Requirements

Each dashboard card must redirect users to the related page:

Dashboard Card	Redirect Page
Total Purchase Cost	Purchase Invoice Page
Total Expenses Cost	Expenses Page
Total In-Stock Amount	Items Page
Total Sale Price	Sale Invoice Page
Total Repair Cost	Repair Invoice Page
Total Sold Inverters Profit	Sold Inverters Page
Total Amount To Paid	Supplier Payments Page
Total Amount To Received	Customer Payments Page
Total Current Balance	Accounts Page
Total Production Cost	Production Page

Profit Formula

Overall Profit shall be calculated as:

Overall Profit = (Total Sale Price + Total Repair Cost + Total In-Stock Amount + Total Amount To Received + Total Current Balance)

- (Total Purchase Cost + Total Expense Cost + Total Amount To Paid)

3.2 Accounts Module

Description

The Accounts module shall manage all business transaction accounts.

Table Columns

The Accounts table shall contain:

- ID
- Name
- Account Type
- Opening Balance
- Current Balance
- Created By

Functional Requirements

Account Sorting

- Accounts must be sorted alphabetically from A-Z.

Total Balance

- The system shall calculate total current balance.
- The total balance shall display at the bottom of the table.

Current Balance Logic

The system shall:

- Allow current balance to become negative.
- Deduct payments from current balance.
- Use opening balance for balance adjustments.

Example Logic

If:

- Current Balance = -30
- Opening Balance added = 30

Then: New Current Balance = 0

Additional Features

The system shall provide:

- Add account
- Edit account
- Delete account
- Add opening balance
- Account-to-account transfer
- Search across all columns
- CSV export
- Pagination

3.3 Supplier Module

Description

The Supplier module shall manage supplier information, purchase history, payments, and statements.

Supplier Table Columns

- ID
- Name
- Opening Balance
- Total Purchase
- Total Paid
- Due/Remaining Balance

Functional Requirements

Supplier Sorting

Suppliers shall be sorted alphabetically from A-Z.

Supplier Details Page

Clicking a supplier name shall open a detailed supplier profile page.

The page shall display:

- Opening Balance
- Total Purchase Amount
- Total Paid Amount
- Outstanding Amount
- Current Balance

Balance Formulas

Current Balance = Opening Balance + Total Purchase Amount

Outstanding Balance = Current Balance - Total Paid Amount

Purchase History

Table Columns

- Invoice #
- Date
- Amount
- View
- Edit
- Delete

Functional Requirements

- Display total purchase amount at the end.
- Sort by latest date first.
- Search all columns.

- Export CSV.
 - Download PDF.
 - Pagination.
-

Supplier Payment History

Table Columns

- Invoice #
- Date
- Amount Paid
- Account
- Notes
- View
- Edit
- Delete

Functional Requirements

- Display total paid amount.
 - Sort latest first.
 - Search all columns.
 - Export CSV.
 - Download PDF.
 - Pagination.
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Supplier Statement

Statement Table Columns

- Invoice #
- Date
- Amount
- Account
- Amount Paid
- Outstanding Balance

Statement Rules

The statement shall combine:

- Purchase invoices
- Supplier payments

Footer Summary

The statement footer shall display:

- Opening Balance

- Total Purchase Amount
- Current Balance
- Total Paid Amount
- Total Outstanding Balance

Additional Features

- Download PDF
 - Pagination
 - Row-level View button
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3.4 Customer Module

Description

The Customer module shall manage customer records, sales, repairs, payments, and statements.

Customer Table Columns

- ID
- Name
- Opening Balance
- Total Sales
- Total Repairs
- Total Payments Received
- Due/Remaining Balance

Customer Detail Page

The page shall display:

- Opening Balance
- Total Sale Amount
- Total Repair Amount
- Total Payment Received
- Outstanding Amount
- Current Balance

Formula

Current Balance = Opening Balance + Total Sale Amount + Repair Amount

Outstanding Balance = Current Balance - Total Payment Received

Sale History

Table Columns

- Invoice #
- Date
- Amount
- View
- Edit
- Delete

Requirements

- Show total sale amount.
 - Date descending order.
 - Search all columns.
 - CSV export.
 - PDF download.
 - Pagination.
-

Repair History

Table Columns

- Invoice #
- Date
- Amount
- View
- Edit
- Delete

Requirements

- Show total repair amount.
 - Date descending order.
 - Search all columns.
 - CSV export.
 - PDF download.
 - Pagination.
-

Customer Payment History

Table Columns

- Invoice #
- Date
- Amount Received
- Account

- Notes
- View
- Edit
- Delete

Requirements

- Show total received amount.
 - Date descending order.
 - Search all columns.
 - CSV export.
 - PDF download.
 - Pagination.
-

Customer Statement

Statement Table Columns

- Invoice #
- Date
- Amount
- Account
- Amount Received
- Outstanding Balance

Statement Sources

The statement shall combine:

- Sales
- Repairs
- Payments

Footer Totals

The statement footer shall display:

- Opening Balance
- Total Sale Amount
- Total Repair Amount
- Current Balance
- Total Received Amount
- Outstanding Balance

Additional Features

- Download PDF
 - Pagination
 - View invoice details
-

3.5 Items Module

Description

The Items module shall manage inventory items and stock valuation.

Top Summary Section

The system shall display:

- Total Stock Value
- Total Units In Stock
- Total Items In Stock

Table Columns

- ID
- Name
- Category
- Type
- Unit
- Average Price
- Total Quantity
- Total Value
- Created By
- Edit
- Delete

Functional Requirements

- Alphabetical sorting.
 - Search all columns.
 - Filters:
 - Raw Material / Final Product
 - In Stock / Out of Stock
 - Categories
 - Export Items PDF.
 - Add New Item.
 - Pagination.
-

3.6 Categories Module

Description

The Categories module shall manage item categories.

Table Columns

- ID
- Name
- Created By
- Edit
- Delete

Functional Requirements

- Search all columns.
 - CSV export.
 - Add category.
 - Pagination.
-

3.7 Recipes Module

Description

Recipes define materials required for inverter manufacturing.

Recipe Logic

The recipe shall:

- Use stock item average prices.
- Calculate total recipe cost.
- Allow additional expense entry.
- Display total recipe value during creation.

Important Rule

Creating recipes shall NOT reduce stock quantity.

Dynamic Price Update

If average item prices change in stock:

- Recipe costs shall automatically update.

Recipe Table Columns

- Final Product
- Recipe Name
- Ingredients Count
- Cost/Unit Price
- Created By
- View

- Edit
- Delete

Features

- Search all columns.
 - CSV export.
 - Add recipe.
 - Total recipe cost display.
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3.8 Production Module

Description

The Production module shall manage inverter manufacturing batches.

Production Process

The user shall: 1. Select recipe 2. Enter production quantity 3. Assign batch number 4. Enter copper amount 5. Select deduction account 6. Enter notes 7. Save production batch

Copper Cost Logic

When production is saved:

- Copper amount shall deduct from selected account.
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Production Batch Table

Columns

- Batch #
- Recipe/Product
- Quantity
- Status
- Cost
- Created On
- Created By
- View
- Edit
- Delete

Functional Requirements

- Search all columns.
- Date descending order.
- Total production cost.

- Total production summary.
- CSV export.

Locked Production Cost Rule

Once production is completed:

- Production cost must remain fixed.
 - Future recipe cost changes must NOT affect previous batches.
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3.9 Purchase Invoice Module

Description

Manage all purchase invoices.

Table Columns

- Invoice #
- Supplier
- Date
- Amount
- View
- Edit
- Delete

Functional Requirements

- Add purchases.
 - Search all columns.
 - CSV export.
 - Date descending order.
 - Total purchase amount.
 - Supplier filter.
 - Date filter.
 - Total purchases summary.
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3.10 Sale Invoice Module

Description

Manage all sales invoices.

Table Columns

- Invoice #
- Customer
- Date
- Amount
- View
- Edit
- Delete

Functional Requirements

- Add sales.
 - Search all columns.
 - CSV export.
 - Date descending order.
 - Total sale amount.
 - Customer filter.
 - Date filter.
 - Total sales summary.
-

3.11 Repair Invoice Module

Description

Manage repair invoices and repair operations.

Table Columns

- Invoice #
- Customer
- Description
- Date
- Amount
- View
- Edit
- Delete

Functional Requirements

- Search all columns.
- CSV export.
- Date descending order.
- Total repair amount.
- Customer filter.
- Date filter.
- FOC/Charged filter.

FOC Repair Logic

If repair is Free of Cost:

- Item quantities must deduct from stock.
- Item value must deduct from stock.
- No amount added to accounts.
- No amount added to repair revenue.

Serial Number Selection

When creating repair invoice:

- Selecting inverter type shall show sold inverter serial numbers.

Example: If user selects 18KW inverter:

- Show serial numbers of sold 18KW inverters.

3.12 Sold Inverters Module

Description

Track all sold inverter units.

Table Columns

- Serial #
- Item Name
- Customer
- Production Cost
- Sale Cost
- Profit

Formula

Profit = Sale Cost - Production Cost

Functional Requirements

- Show totals for:
 - Production Cost
 - Sale Cost
 - Profit
- Total inverter sales summary.

- Date descending order.
 - View details button.
 - Search all columns.
 - Customer filter.
 - Date filter.
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3.13 Supplier Payments Module

Description

Manage supplier payments.

Table Columns

- Invoice #
- Supplier Name
- Date
- Amount
- Created By
- View
- Edit
- Delete

Functional Requirements

- Search all columns.
 - Supplier filter.
 - Date filter.
 - Account filter.
 - CSV export.
 - Add payment.
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3.14 Customer Payments Module

Description

Manage customer payment receipts.

Table Columns

- Invoice #
- Customer Name
- Date
- Amount
- Created By

- View
- Edit
- Delete

Functional Requirements

- Search all columns.
 - Customer filter.
 - Date filter.
 - Account filter.
 - CSV export.
 - Add payment.
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3.15 Stock Adjustment Module

Description

Manage stock corrections and adjustments.

Functional Requirements

The user shall:

- Select item from dropdown.
- Enter quantity.
- Enter unit price.
- Save stock adjustment.

Current Stock Information

The system shall display current stock details for selected item.

Stock Adjustment History Table

Columns

- ID
- Date
- Quantity
- Unit Price
- Reason
- Notes
- Adjusted By

Features

- Adjustment stock button.

- Adjustment history display.
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3.16 Expense Categories Module

Description

Manage expense categories.

Table Columns

- ID
- Name
- Description
- Expenses
- Created By
- View
- Edit
- Delete

Functional Requirements

- Search all columns.
 - CSV export.
 - Alphabetical sorting.
 - Add expense category.
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3.17 Expenses Module

Description

Manage business expenses.

Expense Entry Requirements

While adding expenses:

- System shall calculate total expense live.
- Multiple expense entries shall be supported.

Table Columns

- Date
- Description
- Amount
- Category

- Account
- Created By
- Notes

Functional Requirements

- Display total expense amount.
 - Date descending order.
 - Search all columns.
 - Category filter.
 - Account filter.
 - Date filter.
 - Created by filter.
 - CSV export.
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3.18 User Management Module

Description

Manage user profile information.

User Information Fields

- First Name
- Last Name
- Phone Number
- Email
- Bio
- Address

Functional Requirements

- Edit user information.
- User information shall be used in PDF documents.

PDF Documents Include

- Repair invoices
 - Sale invoices
 - Statements
 - Purchase invoices
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4. Non-Functional Requirements

4.1 Performance Requirements

- System must load pages quickly.
 - Search operations should respond in real-time.
 - Large tables must support pagination.
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4.2 Security Requirements

- Only authorized users may access the system.
 - User actions should be role-controlled.
 - Financial data must be protected.
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4.3 Usability Requirements

- User interface must be easy to understand.
 - Navigation must be simple and user-friendly.
 - Tables and reports must be readable.
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4.4 Reliability Requirements

- Data must remain accurate.
 - Financial calculations must always remain consistent.
 - Production and stock calculations must preserve historical records.
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4.5 Compatibility Requirements

The system should support:

- Desktop browsers
 - Tablets
 - Mobile responsive layouts
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5. Business Rules

1. All monetary calculations must be accurate.
2. Stock quantities cannot become inconsistent.
3. Historical production costs must remain fixed.
4. Search must work across all columns.
5. Financial statements must update automatically.

6. Invoice records must support full tracking.
 7. Repair invoices may be Charged or FOC.
 8. FOC repairs reduce stock only.
 9. Dashboard totals must update dynamically.
 10. Exported PDF and CSV reports must contain accurate information.
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6. Reporting Requirements

The system shall support:

- CSV exports
 - PDF downloads
 - Statements
 - Purchase reports
 - Sales reports
 - Expense reports
 - Payment reports
 - Inventory reports
 - Production reports
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7. Future Enhancements (Optional)

Possible future enhancements include:

- Barcode scanning
 - Mobile application
 - Role-based permissions
 - Notification system
 - SMS integration
 - Email integration
 - Multi-branch management
 - Audit logs
 - Cloud backup
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8. Conclusion

This SRS document defines the complete requirements for the Inventory, Production, Sales, Repairs, and Financial Management System. The application is intended to provide centralized management of inventory, production, accounting, suppliers, customers, expenses, repairs, and reporting.

The system must ensure:

- Accurate financial tracking
- Reliable stock management

- Easy reporting
- Historical data consistency
- Efficient business operations
- User-friendly experience